Lesson plan

Student

Name: **Alexander Botzki**	Subject-didactic cluster: Science and Technology	Academic year: 2025 - 2026

Lesson topic

Strategic use of generative AI for research	Subject: Strategic Use of Generative AI Software

Internship School Internship

Name and address: VIB, Technologiepark 75, 9052 Ghent
Subject mentor: Bruna Piereck

Lessons

Finality/field of study : /			School year: 2025
			Number of students: 17
Date: 15/10/2025	Time: from: 13:30	until : 5:00 PM	Location: GHB Auditorium

Initial situation

	Description of relevant features for this lesson	Concrete actions based on the characteristics recorded in the initial situation
Students	Via survey for the course: participant wishes for recent updates	- I propose some new tools like ASTA, NotebookLM
Class group - group dynamics	- usually a fairly quiet group of adults - encouraging interaction between participants is appropriate	- I provide interactive surveys with Mentimeter - I provide individual exercises and group assignments for 2 or more people - I provide a summary interactive group assignment at the end of the session
Class group – content-related	Through survey for the course	- I am adding an assignment in which the publishers' policies are critically examined
School/situational data	- Participants will be informed by email to set up access to certain software in advance - Participants will work on their own laptop	- if possible before the start of the lesson: check whether wireless internet works on all computers

Commented [A21]: ASTA, NotebookLM and Napkin are unknown to most participants.

Commented [A22]: these questions work well.

Commented [A23]: Participants aren't very keen on collaboration. The room was also very large and many seats were unoccupied, which limited interaction between participants.

Commented [A24]: difficult exercise

Points of interest

Personal points of attention taken from previous lesson(s):	- The part about coding with R is replaced by an interactive group assignment

Sources

Curriculum/notebook: not applicable
(*Visualizing – AI for Students* , 2025) (Andy Stapleton, 2025) (Michigan Institute for Data & AI in Society, 2024) (Knowles, 2024)

Other:

Andy Stapleton (Director). (2025, April 24). *Top 7 Free AI Tools Every Researcher Needs in 2025* [Video recording].https://www.youtube.com/watch?v=vL12MBT1JZw

Knowles, A. M. (2024). Machine-in-the-loop writing: Optimizing the rhetorical load. *Computers and Composition* , *71* , 102826. https://doi.org/10.1016/j.compcom.2024.102826

Lin, Z. (2024). Techniques for supercharging academic writing with generative AI. *Nature Biomedical Engineering* , *9* (4), 426–431. https://doi.org/10.1038/s41551-024-01185-8

Michigan Institute for Data & AI in Society (Director). (2024, May 6). *Generative AI Tutorial Series | AI Tools for Literature Discovery and Summarization | 2024* [Video recording]. https://www.youtube.com/watch?v=5LAbLZUy_SU

Visualizing – AI for Students . (2025, October 4). https://aivoorstudenten.be/en/courses/ai-voor-studenten/lessons/visualiseren/

Appendix n

Presentation:
- https://github.com/vib-tcp/introduction-to-generative-ai/blob/main/presentations/20251015-strategic_use_of_GenAI_Research_2025_V2.5.pptx

Exercises:
- mentimeter https://www.menti.com/alzuad8b2ew8 - slides 45 -

- https://github.com/vib-tcp/introduction-to-generative-ai/exercises
  exercise individual assignment classification of activities writing

- shared google document for participant feedback

- exercise-GenAI-writing-activities.docx

Curriculum objectives/final terms

Other
Select a school year:
Other

Objectives of this course:

- Identify the characteristics, capabilities, and limitations of various generative AI tools used in research workflows.
- Demonstrate the use of generative AI tools to support specific academic tasks, including writing, summarizing, and idea generation.
- Integrate generative AI tools into personal research workflows to increase productivity while maintaining critical control.

LESSONS OBJECTIVES	LEARNING CONTENT	WORKING FORMS/MEDIA/ORGANIZATION	TIME
	Lesson Beginning		
		Working method: relevant context + questions about assessment *Media: Powerpoint (slide 12), mentimeter slides 44-47* *Organization:* The teacher shows slide 12 and asks the following questions. The students provide answers via Mentimeter . **AI in Writing** : *Do you think AI tools should be allowed to help researchers write scientific articles, or does this undermine originality and critical thinking?* • **Ethics and Integrity** : *What ethical challenges do you see in using AI-generated text in scientific publications, and how should journals or institutions deal with this?* Detection **and Reliability** : *AI text detectors often misclassify. How much confidence do you have in these tools, and what would make them more reliable?* • **Governance and Citation Bias** : *If AI tools begin to influence which articles are cited more often, how might this bias affect the governance of scientific knowledge?*	00:00 (13:20) -00:10
	Lesson center		
- The student can list applications of AI in the academic writing process. - The student can compare examples of human- and machine-controlled writing processes. - The student can describe the steps of an AI-supported literature study.	**LESSON PHASE 1: How does generative AI influence the writing process?** - Differences between human writing, human-in-the-loop writing, machine-in-the-loop writing, and synthetic writing. - Related applications of AI in the four categories: assignment feedback, drafting early versions, article summaries, citation formatting, educational materials, language acquisition, thesis supervision. - AI-assisted literature review: develop search strategy, screen results, organize literature, synthesize and revise insights.	*Working method: individual assignment with class discussion, teaching* *Media: Powerpoint (slides 13-17), printouts categories, PostIts* *Organization/Conversation format:* ▪ Before the lesson, the teacher hangs 4 sheets of paper (A3 size, exercise-GenAI-writing-activities.docx) with the four dimensions of the writing process on the wall. ▪ The instructor provides a brief introduction to relevant dimensions of the writing process and briefly explains that Generative AI methods can be relevant in each of these dimensions (slide 13). ▪ The teacher explains the activity and distributes Post-It notes. Slide 14 serves as a reminder during the exercise. ▪ Participants write down at least 4 types of writing activities and stick the PostIts under the given category. ▪ The teacher reviews and discusses the results together with the participants. ▪ The instructor shows slide 15 as a summary of this exercise. The instructor notes that this division of activities may change over the years. ▪ The next slide 16 serves as a transition to go into the steps before the writing process. ▪ The teacher introduces the steps of the AI-supported literature study on slide 17. ▪ Transition: In the next phase, we will look at developing a search strategy using an AI tool.	00:10 - 00:30

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LESSONS OBJECTIVES	LEARNING CONTENT	WORKING FORMS/MEDIA/ORGANIZATION	TIME
- The student can describe how AI tools contribute to developing search strategies for literature research. - The student can generate relevant search terms and search strings for academic research. - The student can formulate and execute a search query using an AI tool and Google Scholar. - The student can analyze and improve the structure of a search query.	**LESSON 2: How do you use Copilot with Google Scholar to develop a literature search strategy?** - Using LLMs for literature research: generating keywords, search strings, and defining research topics. - Illustrating this with three example questions that AI can efficiently answer, such as identifying trends or relevant stakeholders. - Practical application of using search strings in databases like Google Scholar. - Compound search query uses Boolean operators and thematic clustering. - Improve search by, for example, being more specific, using synonyms, or broadening the search context. - the importance of documentation, because every AI query is unique and cannot be reproduced without recording - AI-supported literature research consists of the following phases: search strategy, screening & evaluation, organization, synthesis, revision.	*Working method: teaching, individual assignment on computer, teaching discussion, debriefing/teaching* *Media:* Powerpoint (slides 18–23), Copilot, Google Scholar, shared Google Document *Organization/Conversation format:* ▪ The instructor briefly introduces AI as a way to define search strategies (slide 18). ▪ The teacher explains that AI is useful for discovering unknown concepts and gives three examples of activities on slide 19. ▪ On slide 20, the lecturer explains the use of an AI tool, Copilot, in conjunction with Google Scholar to identify relevant literature sources. ▪ The teacher explains the following exercise. (Individual assignment) Slide 21 serves as a reminder during this exercise. ▪ Participants work individually on this exercise on their computers for about 7 minutes. ▪ The instructor is available for questions during the exercise. ▪ The teacher asks the students about their experience with this approach. ▪ The instructor asks two participants if they can share their Google Scholar query via a shared Google Doc . The instructor displays these two queries and discusses their structure with the participants. ▪ The instructor shows slide 22 as a summary of this exercise. The instructor notes that this query is different for every search in Copilot. Therefore, documentation of the query is important. ▪ The next slide 23 serves as a transition to discuss tools for screening (evaluating, filtering, substantively supplementing) results.	00:30 - 00:50
- The student can identify AI-supported	**LESSON 3: How do you use specific AI tools for exploratory or supplementary literature study?** - AI-supported literature search tools to	*Working method: teaching, individual assignment on computer, teaching discussion/debriefing/teaching* *Media: Powerpoint (slides 24–40), assignment exercises see links* *Organization/Conversation format:*	00:50 - 01:30 break of 10 min

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LESSONS OBJECTIVES	LEARNING CONTENT	WORKING FORMS/MEDIA/ORGANIZATION	TIME
literature search tools. - The student can independently use AI tools for exploratory literature exploration. - The student can explain the added value of network analysis for literature exploration. - The student can compare different AI-based literature search tools based on their functionalities.	complement traditional search methods such as Google Scholar and AI tools added to a library catalog - Tools: JSTOR, Scopus AI, Semantic Scholar - The AI Literature Assistants are positioned within two approaches: via chatbot interaction or via network analysis - Tools: Perplexity, Asta, ResearchRabbit - Students are introduced to tools that support literature exploration via chat (Perplexity, ASTA). - Tools like ResearchRabbit provide visual representations of related publications. - Students complete an individual assignment with ResearchRabbit.	▪ The instructor mentions that in this phase, we supplement the results of the initial Google Scholar search with results from literature databases and AI-supported assistants. (slide 24) ▪ On slide 25, the lecturer gives the example of JSTOR, a tool from university libraries, which now also has an AI-supported search function. ▪ In slide 26, the lecturer explains that specific online providers of AI-supported searches (literature exploration) use "Semantic Scholar" as a corpus. ▪ On slide 27, the instructor then outlines two different approaches to exploring the literature: using a chat function or network analysis starting from core articles. The instructor provides examples. ▪ On slide 28, the instructor lists tools that offer the first approach, "exploration via chatbot." Two tools, Perplexity and ASTA, are then discussed in an individual assignment. ▪ The individual assignment is explained and slide 29 is shown during the assignment (7 minutes). ▪ After the assignment, the students are asked in class what differences they notice between these two tools. ▪ The instructor presents some tools to map and visualize connections between articles in the literature. ▪ Slides 32 to 36 briefly explain how ResearchRabbit works. ▪ Slide 37 explains the individual assignment for the exploratory use of Research Rabbit. Students conduct a search using Research Rabbit and document their process (7 min). ▪ After the assignment, the students are asked in class what qualitative differences they notice between the first two tools and Research Rabbit. ▪ The instructor concludes this lesson phase by providing some tips and tricks for obtaining information via chatbots about similar tools.	
- The student can various types of AI tools for literature summary analyze .	**LESSON PHASE 4: What are the differences in approach and functionality between AI tools that summarize scientific literature?** - Typology of AI tools: summarization tools, document analysis, exploration - introducing AI tools for literature summarization: Elicit, Asta, and NotebookLM	*Working method: teaching, educational discussion, teaching, individual assignment on computer, debriefing/teaching* *Media: Powerpoint (slides 41–61), assignment exercises (see links), mentimeter slide 48* *Organization/Conversation format:* ▪ The teacher provides an overview of tools that can summarize the content of several articles (slide 41). ▪ On slide 42, the instructor indicates that there are three different approaches to summarizing literature. Elicit, Asta, and NotebookLM will be discussed later.	01:30 - 02:15 (15:10 third topic 15:30 Petal end 15:40

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LESSONS OBJECTIVES	LEARNING CONTENT	WORKING FORMS/MEDIA/ORGANIZATION	TIME
- The student can An select suitable tool for summarizing relevant literature . - The student can understand the ethical and analyze legal implications of sharing documents with AI tools . - The student can demonstrate the operation of Elicit and NotebookLM using a concrete assignment .	- distinguish three approaches: tools with own corpus, tools with PDF upload, tools with chat interface - discuss responsibilities for document upload: attention to copyright and licenses - explaining “fair use” and qualifying copyright in the context of AI training - demonstrating Elicit: use of criteria such as summary, methodology, results and limitations - recommend CC-BY as a suitable license for use in AI tools - NotebookLM proposals: working with your own documents and generating mind maps - carrying out assignment with NotebookLM: uploading three CC-BY files, creating a notebook and mind map	▪ The instructor asks what responsibilities the end user has when sharing files with AI tools. "What ethical and legal responsibilities does an end user have when uploading files to AI tools, such as respecting copyright, privacy, and confidentiality?" How can an end-user assess whether a file is suitable and safe to upload to an AI tool, and what steps should be taken to prevent risks such as data leaks or misuse of information?” The instructor mentions that there are some key responsibilities when sharing files with the AI tools for summarizing. ▪ Afterwards, on slide 44, the lecturer proposes “fair use” of protected material and qualifies that we should not be more papal than the Pope, since OpenAI did not take copyright into account in their model training. ▪ In slide 45, the instructor explains that when sharing files, we consider the license granted by the work's owner. CC-BY is the most recommended. ▪ Slides 46, 47 and 48 introduce the Elicit tool, which allows us to summarize a large number of articles according to a list of criteria. ▪ The teacher shows a shared Elicit notebook through a short demonstration. ▪ On slide 50, the teacher presents the various criteria and concludes the discussion of Elicit with an overview of pros and cons (slide 51). ▪ Slides 53 through 56 briefly introduce chatting with your literature tool, Petal. After uploading PDFs, the content can be explored by asking questions via chat . The instructor states: " First verify the proper functioning of such tools with topics the user is familiar with. There's a good chance that such niche applications will be overtaken by general-purpose chatbots like Copilot/ChatGPT or applications like the following interesting tool." ▪ On slides 57 to 59 the operation of NotebookLM is briefly explained and demonstrated with an example (see link on slide 58). ▪ Slide 59 explains the individual assignment for the exploratory use of NoetbookLM. Students conduct their own search using a tool of their choice to download three files with the correct CC-BY license, or use the example files. They enter these files into a new notebook in NotebookLM and create a mind map. They briefly reflect on this exploration method (7 min). ▪ After the assignment, the students are asked via Mentimeter (slide 48) what applications they can imagine for using Elicit and NotebookLM.	

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LESSONS OBJECTIVES	LEARNING CONTENT	WORKING FORMS/MEDIA/ORGANIZATION	TIME
- The student can potential applications of AI tools for your own research practice to frame .		▪ The teacher moves on to the next phase of the lesson, the final steps of the process, synthesis and revision.	
- The student can identify the different dimensions of the writing process analyze (depth , fluency , accuracy). - The student can use the Human-in-the-loop sandwich model for to write describe . - The student can describe the five levels of LLM use in the writing process to separate . identify current guidelines and challenges citation of AI use in scientific to write to analyze . - The student can analyze current guidelines and challenges surrounding citation of AI use in scientific writing.	**LESSON PHASE 5: How do we structure the writing process and which quality dimensions are important?** - Dimensions of the writing process: depth (ideas, sources, analysis, language use), fluency (clear expression, coherence), accuracy (spelling, grammar, citations). - HITL Writing: Exploration with AI tools, Draft by human, AI provides feedback, Human performs quality control and has final responsibility - Five levels of LLM use: 1. Basic editing (spelling, grammar) 2. Structural editing (paraphrasing, translating) 3. Derived content (summarizing, titles) 4. New content (expanding, brainstorming) 5. Evaluation/feedback (judging quality). - No standardized approach across publishers; variation in policies (Ganjavi et al., 2024). - Important aspects of the writing process	*Working method: teaching, group assignment on computer, debriefing/teaching* *Media: Powerpoint (slides 62–71), assignment exercises (see links), mentimeter slide 49* *Organization/Conversation format:* ▪ As a brief introduction, the teacher discusses the three dimensions of the writing process and refers to examples from the morning session. (slide 62) ▪ In slide 63, the instructor describes the HITL sandwich of the writing process. ▪ On slides 64 and 65, the instructor explains the five levels of Generative AI used in the writing process. Sample prompts are also provided for each. ▪ The teacher then asks the students via mentimeter (https://www.menti.com/al1vvbryeb91) which levels they have already used to write an article. ▪ The instructor asks the class (slides 66-67) what examples students can give regarding the use of Generative AI. At what level would you indicate Generative AI use? ▪ After the brief discussion, the lecturer explains during slide 68 that there is no standardized approach from publishers. ▪ Afterwards, the teacher proposes another assignment (10 minutes). Students work in pairs.	02:15 - 02:45

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LESSONS OBJECTIVES	LEARNING CONTENT	WORKING FORMS/MEDIA/ORGANIZATION	TIME
		▪ Afterwards, the teacher discusses the reflections and gives his/her own example for this assignment. ▪ In slide 70, the instructor transitions to the writing process and mentions the additional training offered, including courses on scientific writing, visualization, and open-source drawing programs such as GIMP and Inkscape. ▪ The instructor explains the creative process of creating illustrations with Napkin.ai. ▪ The teacher moves on to the next phase of the lesson, an integration of the approaches proposed thus far.	
- The student can An suitable approach as long as choose use of AI tools for writing a proposal for fundraising . - The student can use relevant AI tools to create tables, illustrations and text.	**LESSON 6: How to create a short, compelling project proposal using AI tools?**	*Working method: group assignment (freely selectable) with classroom discussion* *Media: Powerpoint (slides 72)* *Organization/Conversation format:* ▪ As an integrated exercise, the teacher discusses the final assignment during slide 72. ▪ Afterwards, the teacher proposes another assignment (15 minutes). Students work in groups of 3-5 people. ▪ The teacher guides the groups in the chosen assignment. ▪ The instructor reviews and discusses the results with the participants and asks about their experiences with the tools.	02:45 - 03:15

Lesson Closing

	A summary of the main lesson content via a quiz. The answers are indicated in this document: https://raw.githubusercontent.com/abotzki/presentation/refs/heads/master/quiz-genAI.md	Working method: formative assessment via quiz The instructor projects the quiz and goes over the questions. Media: https://liascript.github.io/course/?https://raw.githubusercontent.com/vib-tcp/introduction-to-generative-ai/refs/heads/main/supplementary/quiz.md	03:15 - 03:30

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